Technical Requirements Document (TRD): LYS Platform V3.0 - Unified Global System

Version: 3.0 (Global Production Ready)
Status: Approved for Final Engineering
Objective: A single, polymorphic SaaS engine that adapts to US (decentralized), International (centralized), and African (hybrid/gatekeeper) educational hierarchies while gamifying the "Climb" toward career success.

1. Core Architecture & Tech Stack

- **Frontend:** Next.js (React) with Tailwind CSS.
- **Backend:** Node.js (NestJS) for scalable, typed microservices.
- **Database:** PostgreSQL (Relational integrity for Authority Trees).
- **Infrastructure:** Multi-region deployment (AWS/Azure) to support **Data Sovereignty** (Africa/EU).

2. The Unified Data Model

2.1 The Global Authority Tree (Polymorphic Hierarchy)

This table replaces separate regional models. The system traverses this tree to determine if it should look for a "Local District" (US) or a "National Ministry" (Kenya/France).

Table: authorities

Column	Type	Description
id	UUID	PK
name	VARCHAR	e.g., "WAEC", "Ministry of Education (Japan)", "Texas TEA"
level	ENUM	supranational, national, regional_state, local_district, school
parent_id	UUID	Self-referencing FK (Builds the pyramid)
model_type	ENUM	bottom_heavy (US), top_down_unitary (Africa/Asia), federal_hybrid (Nigeria/Germany)
residency_region	VARCHAR	Defines where data must be stored (Compliance)

2.2 The LYS Milestone Engine (Being, Knowing, Doing)

Every task in the system is tagged with the LYS methodology and aligned to standards set by the authorities tree.

Table: milestones
| Column | Type | Description |

id	UUID	PK
category	ENUM	Being (Character), Knowing (Standards), Doing (Assignments)
authority_id	UUID	FK -> authorities.id (The source of the requirement)
is_gatekeeper	BOOLEAN	If TRUE, failure triggers "Alternative Path" logic (Africa focus)
is_hard_deadline	BOOLEAN	Prevents the "Bump" algorithm from shifting this date
standard_code	VARCHAR	e.g., "CCSS.ELA-LITERACY.RH.9-10.2"

3. Global Business Logic

3.1 The "Authority-Aware" Roadmap

The app logic switches UI and data constraints based on the model_type:

- **US View:** Users see Local District notifications; high teacher autonomy allows "custom milestones."
- **International/African Unitary View:** The roadmap is "Locked" to the National Ministry Syllabus. Teachers cannot move milestones; they can only track "Coverage."
- **Federal Hybrid View:** Logic splits: Primary Milestones link to Local Authorities; Secondary/Tertiary link to National/Federal Authorities.

3.2 Global Gamification & Weighting (The Matrix)

The LYSAlignmentMatrix calculates progress based on regional value systems:

- **US/OECD Weight:** Focuses on "Personalized Feedback" and "Continuous Progress."
- **African/Asian Weight:** Focuses on "Certification as Currency." A "Distinction" in a high_stakes_gatekeeper assessment (WAEC/Gaokao) provides a 2.5x multiplier to the user's "Career Ready" score.
- **Learning Crisis Guardrail:** In regions with high pupil-to-teacher ratios (e.g., 58:1), the system automatically injects "Peer-to-Peer Learning" milestones to ensure progress continues without 1:1 teacher time.

3.3 Monthly Workforce Data Sync (BLS.GOV & Beyond)

A worker/cron job updates the workforce_trends table monthly to keep the "Career" side of the matrix accurate.

- **US:** Scrapes BLS.GOV and O*NET.
- **Global:** Scrapes OECD PISA rankings and UNESCO education budget trackers (The "20% Benchmark").
- **Impact:** If a nation's PISA scores drop ("PISA Shock"), the system alerts students to "Policy Volatility" in their roadmap.

4. Developer Implementation Guide

4.1 The "Bump" Algorithm (Universal)

Shifts non-gatekeeper tasks.

- **Logic:** IF milestone.is_gatekeeper == FALSE AND milestone.is_hard_deadline == FALSE THEN shift(date, days).
- **Constraint:** Cannot bump past a high_stakes_gatekeeper exam date.

4.2 LYS Matrix Calculation

$$\text{Student Score} = (\sum \text{Completed Weights}) \times \text{Regional Multiplier}$$

- **Regional Multiplier:** Based on teacher_professionalism (OECD data) or certification_value (National Exam performance).

5. Security, Privacy & Sovereignty

1. **US (FERPA/COPPA):** Data access limited to linked guardian_id.
2. **Europe (GDPR):** Explicit consent and right-to-erasure.
3. **Africa (Data Sovereignty):** All data for residency_region: 'NG' or 'KE' must reside on servers within those geographic boundaries as per national law.

6. Phasing

1. **Phase 1:** Core Authority Tree & US Bottom-Heavy Model.
2. **Phase 2:** Gatekeeper Logic & African Unitary Model.
3. **Phase 3:** Monthly External Data Integration (BLS/OECD/WAEC).
4. **Phase 4:** Advanced Gamification (Alignment Matrix V1).